

Radians: A Different Unit for the Measure of An Angle (Homework)

Convert the following angle measures to radian units.

① 40°

② 100°

③ 255°

④ 422°

Convert the following angle measures to degree units.

⑤ $\frac{\pi}{5}$

⑥ $\frac{3\pi}{8}$

⑦ $\frac{9\pi}{10}$

Round to
the nearest
thousandth.

⑧ -4.74

Find the values of the following expressions without a calculator.

⑨ $\sin \frac{\pi}{4}$

⑩ $\cos 2\pi$

⑪ $\tan \frac{\pi}{6}$

⑫ $\csc \frac{3\pi}{4}$

⑬ $\sec \frac{2\pi}{3}$

⑭ $\cot\left(-\frac{\pi}{2}\right)$

Find the value of each expression using a calculator.
Round to the nearest thousandth.

(15) $\sin \frac{\pi}{10}$

(16) $\cos \frac{3\pi}{8}$

(17) $\tan(-2)$

Find θ if $0 \leq \theta \leq 2\pi$. Write your answers in
radian units.

(18) $10\sin\theta - 4\sin\theta = 2 + 2\sin\theta$

(19) $\tan\theta - 3\tan\theta = 0$ (20) $3\csc\theta = -4\sqrt{2} - \csc\theta$

Answers

(1) $\frac{2\pi}{9}$ (6) 67.5° (11) $\frac{\sqrt{3}}{3}$ (16) 0.383

(2) $\frac{5\pi}{9}$ (7) 162° (12) $\sqrt{2}$ (17) 2.185

(3) $\frac{17\pi}{12}$ (8) -271.582° (13) -2 (18) $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$

(4) $\frac{21\pi}{90}$ (9) $\frac{\sqrt{2}}{2}$ (14) 0 (19) $\theta = 0, \pi, 2\pi$

(5) 36° (10) 1 (15) 0.309 (20) $\theta = \frac{5\pi}{4}, \frac{7\pi}{4}$